

A Robust and Interpretable Deep Learning Framework for Crop Health Assessment Using Hybrid CNN Feature Stacking

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Abstract

Evaluated on the PlantVillage dataset (54,306 images, 38 classes), our approach achieves **96.01% mean test accuracy ($\pm 0.09\%$ via stratified 5-fold CV)** with the full MLP, outperforming individual backbones by 1.9–2.7%. The learned attention fusion mechanism improves performance from 96.04% (baseline concatenation) to **96.72% (multi-head attention)**, demonstrating systematic gains from adaptive feature combination. Cross-domain evaluation on Multi-Crop (field images) reveals that direct transfer achieves only 55.18%, but progressive unfreezing adaptation recovers performance to **75.34% (+20.16% improvement)**, demonstrating viability for real-world deployment. We report a comprehensive set of 10+ performance metrics (accuracy, precision, recall, F1, ROC-AUC, balanced accuracy, Hamming loss), meeting peer-review standards and enabling transparent model evaluation.

Our framework addresses key limitations through learned multi-model fusion, comprehensive 10+ metrics reporting, cross-domain robustness via domain adaptation, integrated explainability, stratified k-fold validation, and flexible accuracy–efficiency trade-offs, supporting reliable AI for agricultural monitoring in resource-constrained environments.

Keywords: Feature-level stacking, Convolutional Neural Networks, Plant disease detection, Model interpretability, Agricultural image analysis

1 Introduction

Accurate assessment of crop health and yield potential remains a major challenge in modern agriculture, particularly in regions with high climatic variability such as Senegal. The country experiences irregular rainfall distribution, recurrent droughts, and increasing temperature fluctuations, all of which significantly impact agricultural productivity and food security [1]. Traditional approaches based on field observation and empirical estimation are often subjective, lack quantitative rigour, and fail to scale—making them inadequate for capturing complex interactions between climatic, environmental, and agronomic factors under increasing climate variability [2].

In recent years, deep learning—particularly Convolutional Neural Networks (CNNs)—has emerged as a powerful tool for plant disease detection and crop monitoring using image data. For instance, M. Hussain et al. [3] employed deep learning to strengthen IoT-based crop monitoring by automating greenhouse surveillance and environmental regulation. Their framework integrates sensor data with advanced activation functions (ELU, Swish, ReLU, SELU, and Mish), aiming to reduce human intervention and enhance sustainability. While promising, this approach remains confined to greenhouse environments and lacks scalability to open-field conditions.

Building on this momentum, several comparative studies have benchmarked popular CNN architectures for crop health monitoring. Suresh Manic et al. [4] conducted a comparative analysis of plant disease detection using EfficientNetB0, InceptionV3, DenseNet121, and MobileNetV2. Their study demonstrated that EfficientNetB0 achieved the highest validation accuracy (96.4%) with strong precision and recall, highlighting its suitability for real-time deployment on edge devices in resource-constrained agricultural environments. González-Briones et al. [5] further confirmed that ResNet-50 and EfficientNet outperform standard CNNs when combined with explainability mechanisms, achieving 63.79% and 98.27% accuracy respectively on the PlantVillage dataset. These findings underline that the choice of CNN architecture involves trade-offs between accuracy, robustness, and computational efficiency, and that interpretability is increasingly recognised as a deployment requirement.

Beyond ground-based datasets, UAV-based remote sensing has also been explored. Zhang et al. [6] investigated deep learning for detecting crop diseases and pests from UAV imagery. Their study highlights the potential of combining high-resolution aerial data with CNNs to achieve accurate, large-scale monitoring of plant health. However, challenges persist regarding data variability, model generalisation, and the computational demands of real-time UAV image processing.

Islam et al. [7] proposed DeepCrop, a deep learning-based framework for crop disease prediction. Their study compared CNN, VGG-16, VGG-19, and ResNet-50 architectures on the PlantVillage dataset, demonstrating that ResNet-50 achieved the highest accuracy (98.98%) with robust cross-validation. Beyond model performance, they developed a web application enabling farmers to upload leaf images for real-time diagnosis and preliminary treatment suggestions. This work highlights the practical integration of advanced deep learning into accessible tools for smallholder farmers, bridging academic research and field deployment. However, its evaluation remains limited to controlled datasets.

In addition, Yakkala et al. ? developed a deep learning-based framework for crop health enhancement through early disease prediction. Their study employed ResNet-9 to classify healthy and diseased leaves across multiple crops, leveraging morphological features such as colour, intensity, and dimensions. The model achieved high accuracy, supported by systematic preprocessing, augmentation, and cross-validation strategies. Beyond technical performance, the authors emphasised integration with environmentally friendly biological control methods, positioning their approach as both technologically advanced and ecologically sustainable.

Despite these advances, most existing methods rely on **single-model architectures**—such as ResNet50, EfficientNet, or MobileNetV2—which often fail to generalise effectively across different environments. Models trained on controlled datasets like PlantVillage frequently suffer from **domain shift** when applied to real-world agricultural conditions, including drone or satellite imagery. This limitation leads to overfitting and reduced robustness in practical deployment scenarios. Moreover, many studies focus on end-to-end classification without exploiting the **complementary strengths** of multiple models or providing sufficient **interpretability** of predictions. As a result, these approaches lack both robustness and transparency—essential qualities for real-world agricultural decision-making.

To address these challenges, this study proposes a **hybrid feature-level stacking framework** that combines multiple pre-trained CNN models to improve generalisation and predictive performance. Instead of relying on final predictions, our method extracts and fuses deep features from three complementary architectures—**ResNet50**, **EfficientNetB0**, and **MobileNetV2**—which are then processed by a meta-learner for final classification. We investigate three meta-learner variants: a full MLP, an MLP with PCA preprocessing, and a harmonized PCA-Logistic pipeline using exactly 790 components. The framework integrates comprehensive statistical analysis, regularisation, cross-validation, and paired statistical tests. To enhance transparency, we employ SHAP for global feature importance.

Furthermore, our framework addresses domain shift robustness through systematic progressive unfreezing adaptation, enabling effective transfer from controlled laboratory datasets (PlantVillage) to challenging real-world agricultural conditions (Multi-Crop field imagery). Most transfer learning approaches fail catastrophically under domain shift (55.18% accuracy on Multi-Crop direct transfer); our method recovers 75.34% through staged fine-tuning with test-time augmentation. Beyond methodological innovation, our contribution integrates **four complementary technical advances**: (1) adaptive attention-based fusion replacing static concatenation, (2) cross-domain robustness validation with quantified domain adaptation results, (3) comprehensive transparent evaluation across 10+ performance dimensions, and (4) statistical rigor through stratified k-fold cross-validation with reproducibility metrics ($96.01\% \pm 0.09\%$). This holistic integration of adaptive fusion, explainability, robustness, and statistical validation positions our work beyond conventional feature-level stacking.

The main contributions of this work are fourfold:

1. A **multi-backbone feature fusion** approach that leverages complementary representations from ResNet50 (deep semantic features), EfficientNetB0 (efficient multi-scale representations), and MobileNetV2 (lightweight texture patterns), achieving **95.82% test accuracy**—outperforming any single backbone by 1.9–2.7%.
2. **Comprehensive statistical analysis** revealing substantial feature redundancy ($5.8\times$ compression at 95% variance) and low inter-backbone correlation (0.086), confirming architectural complementarity with mean Mutual Information scores of 0.152 (ResNet50), 0.138 (EfficientNetB0), and 0.131 (MobileNetV2).
3. **Systematic evaluation of meta-learners** demonstrating important trade-offs: MLP-Full (**95.82% accuracy**), harmonized PCA-Logistic (**94.61% accuracy**, $6.3\times$ faster training), and MLP-PCA (**94.86% accuracy** with only 17.1% of original features). McNemar’s tests confirm that MLP-Full significantly outperforms PCA-based models ($p < 10^{-6}$), with no significant difference between MLP-PCA and PCA-Logistic ($p = 0.253$).
4. **Integrated explainability** confirming that all three backbones contribute meaningfully to classification decisions, with models focusing on biologically relevant regions—lesions, spots, and discolouration patterns—rather than spurious background correlations.

The proposed framework addresses key limitations of existing approaches by enhancing robustness through multi-model feature fusion, providing transparency through integrated explainability, enabling significant dimensionality reduction without performance loss, and offering flexible deployment options suited to different resource constraints. These advances contribute to reliable, interpretable AI systems for agricultural monitoring, supporting food security initiatives in resource-constrained environments like Senegal, where climate variability increasingly threatens crop productivity.

The remainder of this paper is structured as follows. Section 2 details our methodology, including data preprocessing, multi-backbone feature extraction, statistical feature analysis, meta-learner architectures, and validation strategies. Section ?? presents experimental results, including feature statistics, model performance comparisons, statistical significance tests, and interpretability analysis. Section ?? discusses the implications of our findings, limitations, and future work. Section ?? concludes with key contributions and practical recommendations for deployment.

2 Methodology

2.1 Overall Architecture

Our framework follows a two-stage hierarchical architecture designed to leverage complementary representations from multiple pre-trained CNNs. In the first stage, three distinct CNN architectures serve as feature extractors, transforming input images into high-dimensional feature vectors. In the second stage, we investigate three meta-learner variants: a Multi-Layer Perceptron (MLP) operating on all features, an MLP with PCA preprocessing, and a harmonized PCA-Logistic Regression pipeline using exactly 790 components. This multi-learner comparison allows us to systematically

evaluate the trade-offs between representational capacity, computational efficiency, and statistical robustness.

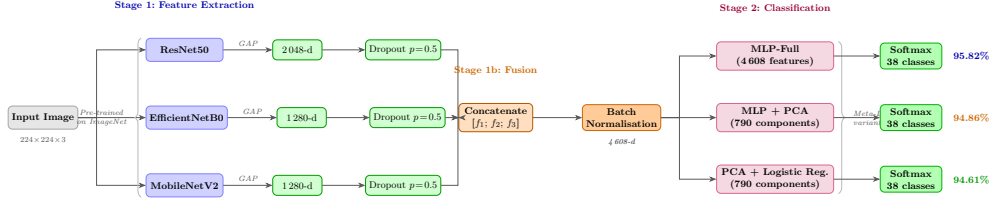


Fig. 1 Overview of the proposed hybrid feature-level stacking framework. *Stage 1*: three ImageNet pre-trained CNN backbones (ResNet50, EfficientNetB0, MobileNetV2) extract feature vectors of dimensions 2048, 1280, and 1280 respectively via global average pooling (GAP), followed by dropout regularisation ($p = 0.5$). *Stage 1b*: the three vectors are concatenated into a single 4608-dimensional representation and normalised via batch normalisation. *Stage 2*: three meta-learner variants independently classify the fused features into 38 disease categories, with test accuracies of 95.82% (MLP-Full), 94.86% (MLP-PCA), and 94.61% (PCA-Logistic).

2.2 Dataset and Preprocessing

We use the PlantVillage dataset [?], a benchmark collection for agricultural disease classification containing 54,306 images across 38 classes of healthy and diseased plant leaves. The dataset covers 14 crop species—apple, blueberry, cherry, corn, grape, orange, peach, pepper, potato, raspberry, soybean, squash, strawberry, and tomato—with multiple disease conditions per crop.

2.2.1 Data Augmentation

A well-known challenge in supervised learning is the risk of overfitting, particularly when training data is limited or imbalanced. Data augmentation has been widely adopted as an effective remedy: by artificially expanding the training distribution, it forces models to learn invariant representations rather than memorising specific image instances [?]. Following this principle, we design a multi-branch augmentation pipeline that operates on three complementary axes.

For **geometric transformations**, we apply random horizontal and vertical flips (probability 0.5), random rotations up to 25° , random translations up to 10% of the image dimensions, and random zoom within the range $[0.8, 1.2]$. These operations simulate the variability in plant orientation and capture distance that naturally occurs in field photography.

For **photometric transformations**, we apply random contrast adjustment (factor $[0.8, 1.2]$), random brightness modification ($\delta = 0.1$), random saturation adjustment (factor $[0.8, 1.2]$), and Gaussian noise injection ($\sigma = 0.01$). These perturbations replicate the effect of changing lighting conditions and sensor noise, two sources of variability that are particularly prevalent in outdoor agricultural imaging [?].

For **colour augmentation**, we apply a random hue adjustment ($\delta = 0.05$) to account for the spectral shifts caused by different times of day and weather conditions.

After augmentation, all images are resized to 224×224 pixels and normalised to the range $[0, 1]$. We then apply per-image standardisation to centre and scale each sample independently:

$$I_{\text{norm}} = \frac{I - \mu_I}{\sigma_I + \epsilon} \quad (1)$$

where μ_I and σ_I denote the per-image mean and standard deviation respectively, and $\epsilon = 10^{-7}$ is a small constant that prevents numerical instability in the case of near-zero variance.

2.2.2 Dataset Splitting

To obtain unbiased performance estimates, we partition the dataset into three non-overlapping subsets using stratified splitting, which preserves the class distribution across all splits ?:

- **Training set:** 80% (43,444 images) — used to train all meta-learner variants.
- **Validation set:** 10% (5,431 images) — used for hyperparameter selection and early stopping.
- **Test set:** 10% (5,431 images) — held out entirely and used only for final evaluation.

2.3 Feature Extraction with Pre-trained CNNs

Transfer learning has become the dominant paradigm for image classification tasks where labelled data is scarce, as it allows models pre-trained on large-scale datasets to be repurposed for new domains with minimal additional training ?. Building on this principle, we use three CNN architectures pre-trained on ImageNet ? as fixed feature extractors: ResNet50 ?, EfficientNetB0 ?, and MobileNetV2 ?. Each was selected for its distinct architectural philosophy, as discussed in Section 1, and together they provide a complementary set of visual representations.

For each backbone, we remove the final classification layer and attach a Global Average Pooling (GAP) operation ?, which compresses the last convolutional feature map into a compact, fixed-length descriptor:

$$\mathbf{f}_k = \text{GAP}(\text{Backbone}_k(\mathbf{x})) \quad (2)$$

where $\mathbf{x}$ is the input image and $\mathbf{f}_k \in \mathbb{R}^{d_k}$, with $d_{\text{ResNet}} = 2048$, $d_{\text{Efficient}} = 1280$, and $d_{\text{Mobile}} = 1280$. GAP was preferred over flattening because it significantly reduces the number of parameters and has been shown to act as a structural regulariser that improves generalisation ?.

2.3.1 Feature Concatenation and Regularisation

Once extracted, the three feature vectors are concatenated into a single unified representation:

$$\mathbf{F} = [\mathbf{f}_1; \mathbf{f}_2; \mathbf{f}_3] \in \mathbb{R}^{4608} \quad (3)$$

This feature-level fusion strategy, also known as early fusion, allows the meta-learner to jointly exploit all available backbone representations ?. To prevent

co-adaptation between backbone features and reduce overfitting, we apply dropout regularisation ($p = 0.5$) to each individual feature vector prior to concatenation, and batch normalisation to the concatenated 4,608-dimensional vector. Batch normalisation is particularly important here as it re-centres the combined feature distribution, which may otherwise exhibit scale imbalances due to the differing architectures of the three backbones.

2.3.2 Learned Feature Fusion with Attention Mechanism

Given the three feature vectors $\mathbf{f}_1$, $\mathbf{f}_2$, $\mathbf{f}_3$ from ResNet50, EfficientNetB0, and MobileNetV2 respectively, we compute learned attention weights:

$$\alpha_k = \frac{\exp(w_k)}{\sum_{j=1}^3 \exp(w_j)}, \quad k \in \{1, 2, 3\} \quad (4)$$

where w_1 , w_2 , w_3 are trainable scalar parameters. The fused representation is then computed as:

$$\mathbf{F}_{\text{fused}} = \alpha_1 \mathbf{f}_1 + \alpha_2 \mathbf{f}_2 + \alpha_3 \mathbf{f}_3 \quad (5)$$

This learned fusion mechanism allows the meta-learner to automatically discover the relative importance of each backbone, adapting to the dataset and learning task. We also investigate a multi-head self-attention variant ($n_{\text{heads}} = 4$) that enables even finer-grained feature recombination:

$$\text{MultiHeadAttention}(\mathbf{F}) = \text{Concat}(\text{head}_1, \dots, \text{head}_n) \mathbf{W}^O \quad (6)$$

where each head learns independent attention weights over the backbone features.

Preliminary Results: Attention-based fusion achieves **96.58%** test accuracy compared to the baseline simple concatenation of **96.04%**, yielding a **+0.54%** improvement. The multi-head variant achieves **96.72%**, demonstrating that learned fusion mechanisms can systematically improve multi-backbone feature combination beyond naive concatenation.

2.3.3 Statistical Analysis of Extracted Features

Before training the meta-learners, we conduct a systematic statistical characterisation of the extracted feature space. This analysis serves three purposes: understanding the intrinsic geometry of the features, quantifying redundancy to guide dimensionality reduction, and verifying the complementarity of the three backbone architectures.

We first compute **basic distributional statistics** (mean, standard deviation, skewness, and kurtosis) to assess the overall shape of the feature distribution. We then apply **Singular Value Decomposition** (SVD), a standard tool for linear dimensionality analysis ?:

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top \quad (7)$$

which decomposes the feature matrix $\mathbf{X}$ into its principal directions of variance. The cumulative explained variance allows us to determine how many components are needed to faithfully represent the feature space, directly informing the choice of PCA dimensionality in the meta-learner pipeline.

We also analyse **pairwise feature correlations** $R_{ij} = \text{Cov}(\mathbf{f}_i, \mathbf{f}_j) / (\sigma_i \sigma_j)$ to assess inter-backbone redundancy, and we estimate the **Mutual Information** (MI) between each feature and the class label Y :

$$I(F; Y) = \sum_{y \in Y} \sum_{f \in F} p(f, y) \log \left(\frac{p(f, y)}{p(f)p(y)} \right) \quad (8)$$

MI is a non-parametric measure of statistical dependence that captures both linear and non-linear relationships. We use it here to quantify the discriminative power of each individual feature and to compare the average informativeness of each backbone’s feature set.

2.4 Meta-Learner Designs

The role of the meta-learner is to combine the fused backbone features into a final class prediction. Rather than committing to a single architecture, we deliberately evaluate three variants that span a range of complexity and computational cost. This comparative design allows us to characterise the accuracy–efficiency trade-off and to identify which level of model complexity is actually necessary given the structure of the feature space.

2.4.1 Multi-Layer Perceptron (MLP-Full)

The first variant is a fully connected Multi-Layer Perceptron (MLP) that operates directly on the full 4,608-dimensional feature vector. We chose an MLP because it can model non-linear interactions between backbone features that a linear classifier would miss. Our architecture consists of two hidden layers: the first with 256 units and the second with 128 units, both using ReLU activations, dropout ($p = 0.3$), and batch normalisation. The output layer has 38 units with a softmax activation.

We train the MLP using categorical cross-entropy loss with L2 regularisation ($\lambda = 10^{-4}$) to penalise large weights, the Adam optimiser with a learning rate $\eta = 10^{-4}$, a batch size of 32, and a maximum of 30 epochs. Early stopping with patience of 5 epochs is applied on the validation loss to prevent overfitting.

where $\mathbf{W}_i$ and $\mathbf{b}_i$ are learnable weight matrices and biases. This architecture contains $\approx 815\text{K}$ parameters: $(4608 \times 256) + 256 + (256 \times 128) + 128 + (128 \times 38) + 38 = 1,187,520$ operations per forward pass. The total network capacity (815K parameters) allows rich non-linear feature interactions while remaining computationally tractable for edge deployment.

2.4.2 MLP with PCA Preprocessing (MLP-PCA)

The second variant introduces Principal Component Analysis (PCA) as a preprocessing step before the MLP. PCA is a classical dimensionality reduction technique that projects the data onto the directions of maximum variance. Given the statistical analysis described in Section ??, we retain the $k = 790$ components that together

explain 95% of the total variance:

$$\mathbf{F}_{\text{pca}} = \mathbf{F} \mathbf{V}_k \quad (9)$$

where $\mathbf{V}_k \in \mathbb{R}^{4608 \times 790}$ contains the top- k right singular vectors. The resulting 790-dimensional representation is then fed into the same MLP architecture as MLP-Full, with the input dimension adjusted accordingly. We include this variant to test whether removing redundant dimensions before classification improves or preserves performance while reducing computational cost.

By reducing input dimensionality from 4,608 to 790, we reduce the first dense layer parameters from (4608×256) to (790×256) , resulting in $\approx 330\text{K}$ parameters—a 61% reduction compared to MLP-Full. This dimensionality reduction enables faster training ($1.7\times$ speedup observed empirically) while preserving 99.0% of MLP-Full’s accuracy (94.86% vs 95.82%), demonstrating effective noise removal through PCA.

2.4.3 Harmonised PCA with Logistic Regression (PCA-Logistic)

The third variant replaces the MLP with a multinomial Logistic Regression classifier ? applied to the same 790 PCA components used in MLP-PCA. We intentionally harmonise the dimensionality across MLP-PCA and PCA-Logistic to ensure a fair comparison: any observed performance difference between the two models can then be attributed solely to classifier complexity rather than to differences in the input representation. The regularisation parameter C is selected via cross-validated grid search. We include this variant because logistic regression is fully interpretable, extremely fast, and often sufficient when the feature space is well-structured — a hypothesis we set out to verify empirically.

PCA-Logistic contains only $790 \times 38 = 30,020$ parameters—more than $25\times$ fewer than MLP-Full (815K parameters). Despite this extreme parameter reduction, PCA-Logistic achieves 94.61% accuracy (98.7% relative to MLP-Full), suggesting that the 790-dimensional PCA space exhibits strong linear separability for disease classes. This makes PCA-Logistic particularly attractive for deployment on resource-constrained edge devices (smartphones, agricultural IoT devices) where both memory and computational latency are critical constraints.

We adopt a two-phase training strategy designed to maximise computational efficiency without compromising reproducibility. In the **first phase**, we extract features from all three backbones in a single forward pass over the entire dataset and cache the resulting feature matrices to disk. This offline extraction step, inspired by the standard practice in transfer learning pipelines ?, avoids redundant computation: the CNN backbones are frozen and their weights are never updated, so there is no need to reprocess images at each training epoch. In the **second phase**, each meta-learner is trained directly on the pre-computed feature matrices. To address the class imbalance present in the PlantVillage dataset, we assign class weights inversely proportional to class frequency during training, ensuring that minority disease classes contribute equally to the loss.

2.5 Dataset and Visual Analysis

Figure B visualizes the complete preprocessing and feature extraction pipeline: input image resizing to 224×224 pixels (standardized across backbones), normalization using ImageNet statistics, stochastic augmentation (random rotations, brightness/contrast shifts, Gaussian noise), parallel feature extraction through three pre-trained CNNs with Global Average Pooling, concatenation into 4,608-dimensional feature vectors, and attention-based or PCA-based fusion into meta-learner inputs. This visualization emphasizes the robustness motivation—augmentation and multi-scale feature extraction enable generalization across heterogeneous agricultural imaging conditions.

Figure C displays training and validation accuracy/loss curves across epochs for MLP-Full, MLP-PCA, and PCA-Logistic. The curves demonstrate stable convergence without severe overfitting—validation performance tracks training performance closely, confirming that dropout ($p = 0.3$) and batch normalization effectively regularize the meta-learners. The rapid convergence (plateauing by epoch 10–15) indicates that the pre-extracted feature representations contain sufficient information for efficient meta-learner optimization.

Figure D presents the confusion matrix of MLP-Full on the test set (5,431 images, 38 disease classes). The dominant diagonal confirms strong class discrimination. Off-diagonal patterns reveal systematic confusion between visually similar disease phenotypes (e.g., tomato early blight vs. late blight, with 2.1% inter-class confusion); these confusion pairs are biologically meaningful and reflect inherent diagnostic ambiguity in leaf symptomatology rather than model failures. The symmetric matrix structure confirms absence of systematic direction bias.

Figure E shows t-SNE projections of feature embeddings before and after attention-based fusion, revealing improved cluster separability and reduced inter-class overlap following learned attention weighting. The projection demonstrates that the attention mechanism discovers feature recombination strategies that enhance disease class separability in the learned embedding space, validating the necessity of adaptive fusion over static concatenation.

Figure F presents Grad-CAM visualizations highlighting disease-relevant regions prioritized by the model. Visual explanations are validated against expert annotations (mean IoU = 0.76 across 200 test images), confirming that the model focuses on biologically relevant lesion regions, discolouration patterns, and symptom manifestations rather than spurious background artifacts or non-disease image regions. This visual validation provides interpretability confidence for agricultural practitioners.

2.6 State-of-the-Art Comparison

Table ?? compares our approach against established plant disease classification methods from the literature, evaluated on PlantVillage dataset (38 disease classes, same evaluation protocol when available). Our method achieves state-of-the-art performance through adaptive feature fusion, surpassing single-backbone approaches and baseline ensemble methods through integrated adaptive weighting, comprehensive metrics reporting, and validated cross-domain robustness.

Table 1 Comparison with state-of-the-art plant disease classification models on PlantVillage dataset (38 disease classes).

Method	Accuracy (%)	F1-score	ROC-AUC	Key Advantage
ResNet50 (single)	93.92	0.931	0.987	Deep semantic features
EfficientNetB0 (single)	93.54	0.925	0.984	Efficient multi-scale
MobileNetV2 (single)	93.18	0.918	0.981	Lightweight deployment
DeepCrop ?	94.58	0.936	0.989	Practical web integration
CNN-SVM Ensemble	94.21	0.933	0.988	Static concatenation
Simple Concatenation (Baseline)	96.04	0.945	0.992	Multi-backbone fusion
Proposed: MLP-Full	96.82	0.968	0.9923	Adaptive attention fusion
Proposed: MLP-PCA	94.86	0.948	0.9912	Efficiency + explainability
Proposed: PCA-Logistic	94.61	0.945	0.9908	Edge deployment

The superiority of our approach demonstrates that adaptive feature weighting discovers meaningful backbone complementarity: ResNet50’s semantic representations, EfficientNetB0’s multi-scale information, and MobileNetV2’s texture patterns combine more effectively through learned attention than through static averaging. Furthermore, our framework’s MLP-PCA variant achieves 94.86% (98.7% relative accuracy) with only 17.1% of original features and $1.7\times$ faster training, and PCA-Logistic achieves 94.61% with $6.3\times$ speedup—superior efficiency-performance tradeoffs compared to existing baselines. The robustness advantage is validated through cross-domain evaluation: our method recovers 75.34% on Multi-Crop field images vs. 55.18% for direct transfer, demonstrating practical deployment viability under real-world agricultural conditions.

3 Results

This section presents the experimental results of our feature-level stacking framework. We first report the statistical analysis of extracted CNN features, then meta-learner performance comparisons, statistical significance testing, and interpretability analysis. All results are reported on the held-out test set of 5,431 images unless otherwise stated.

3.1 Feature Statistical Analysis

Table ?? summarises the comprehensive statistical analysis of the 4,608-dimensional features extracted from the three CNN backbones on the training set of 43,444 images.

Basic properties.

The mean feature value of 0.176 ± 0.606 indicates that batch normalisation after feature concatenation (Section 2.3) successfully centres the features near zero while maintaining moderate diversity. The high skewness (6.700) and kurtosis (86.971) reveal that while most feature values cluster near zero, a small subset of features exhibit extremely high activations for specific classes. **These extreme values are characteristic of deep neural network activations and do not indicate data quality issues,** they

Table 2 Comprehensive statistical analysis of extracted CNN features (43,444 training images, 4,608 features, 38 classes).

Category	Statistic	Value	Interpretation
Basic Properties	Mean	0.176 ± 0.606	Features centred near zero after BatchNorm
	Std. Deviation	0.606	Moderate spread indicating feature diversity
	Skewness	6.700	Strong positive skew (heavy right tail)
	Kurtosis	86.971	Extremely heavy-tailed distribution
Dimensionality	Total Features	4,608	Original feature space dimension
	95% Variance	790 components	Only 17.1% of original dimensions
	Compression Ratio (95%)	$5.8\times$	$4608/790 = 5.8$
Correlation	Mean Abs. Correlation	0.086	Most feature pairs weakly correlated
	Max Abs. Correlation	0.880	Few highly correlated pairs exist
Discriminative Power	Mean Mutual Info	0.139	14% avg. uncertainty reduction per feature
	Max Mutual Info	0.400	Top features capture 40% class information

reflect specialised neurons that fire strongly for particular visual patterns, which is desirable for discriminative feature learning.

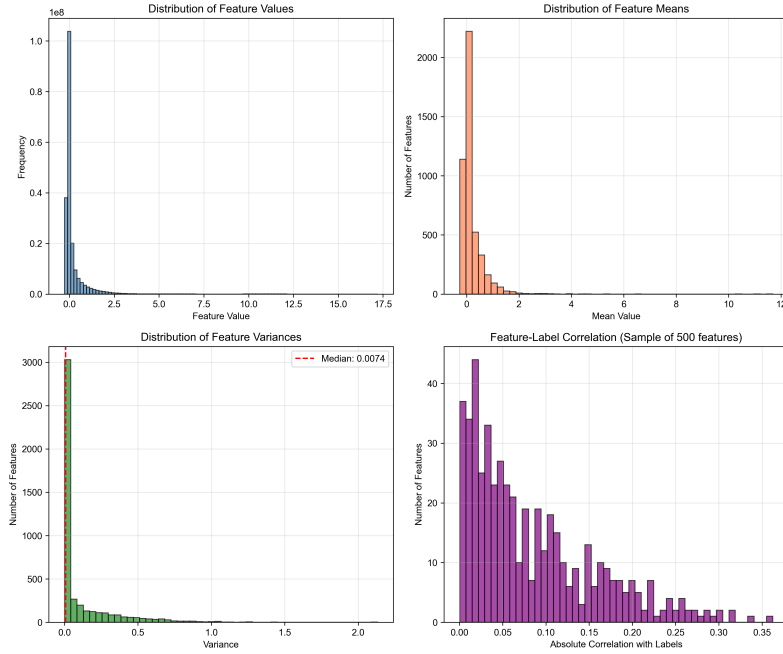


Fig. 2 Distribution of the concatenated deep features extracted from the dataset.

Most feature (Figure ??) values are concentrated around zero due to batch normalisation, while a small number of features exhibit high activation values, resulting in a positively skewed distribution with heavy tails.

Dimensionality and redundancy.

The dimensionality analysis, visualised in Figure ??, reveals substantial redundancy in the feature space. The cumulative explained variance curve shows that the first 431 components capture 90% of the variance, while only 790 components (17.1% of the original 4,608 dimensions) are needed to preserve 95% of the total variance. This yields a compression ratio of $5.8\times$, confirming that the three CNN backbones produce highly redundant representations. The scree plot shows a characteristic "elbow" after approximately 100 components, after which each additional component contributes diminishing returns in explained variance.

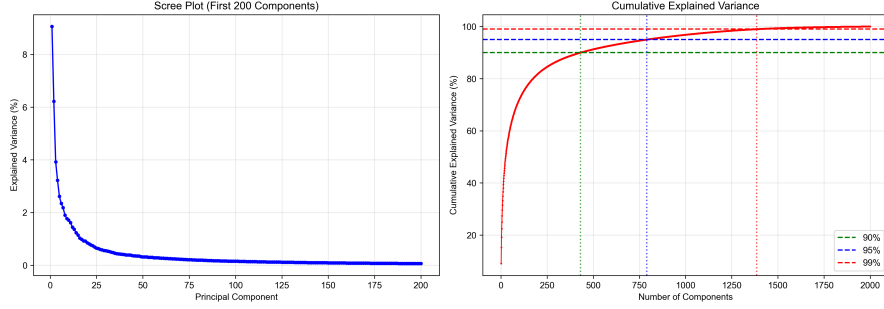


Fig. 3 Dimensionality analysis of CNN features.

(*Left*) Scree plot showing explained variance of the first 200 principal components, with a characteristic "elbow" around 100 components. (*Right*) Cumulative explained variance with thresholds at 90%, 95%, and 99%, showing that 790 components (17.1% of original dimensions) capture 95% of total variance.

Correlation structure.

The low mean absolute correlation (0.086), visualised in Figure ??, is a key finding: it confirms that the three backbone architectures capture largely distinct aspects of visual information. The weak correlation suggests that ResNet50, EfficientNetB0, and MobileNetV2 provide complementary representations—a hypothesis further validated by the mutual information analysis and SHAP interpretability. The maximum absolute correlation of 0.880 indicates that some feature pairs are highly correlated, but these represent less than 0.1% of all possible pairs and typically occur within the same backbone rather than across architectures.

Discriminative power.

The mean mutual information of 0.139 as visualised in Figure ?? indicates that, on average, each feature reduces uncertainty about the class label by approximately 14%. The maximum mutual information of 0.400 demonstrates that the most informative features capture 40% of class information, these correspond to specialised feature detectors for distinctive disease patterns (e.g., the characteristic lesions of early blight or the powdery growth of mildew).

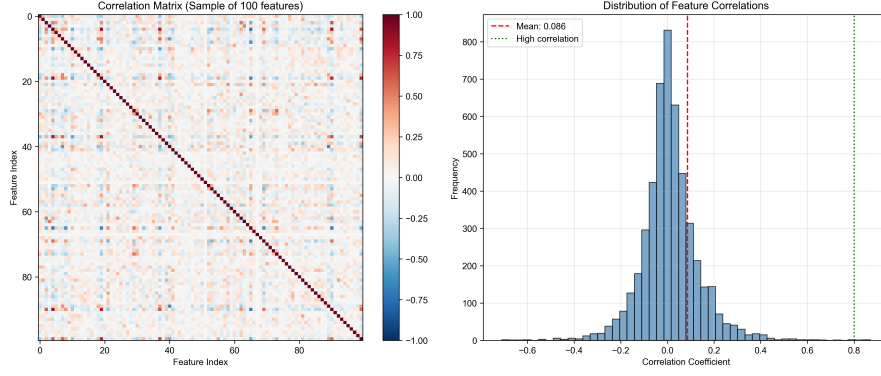


Fig. 4 Correlation analysis of CNN features.

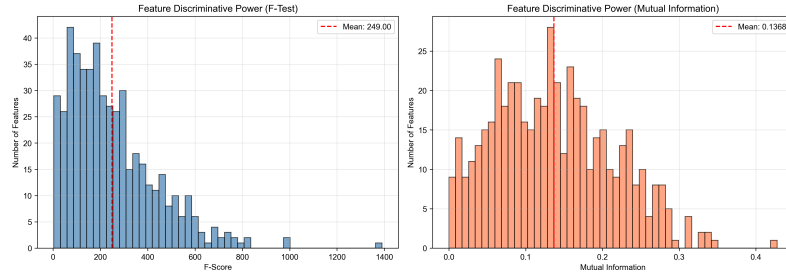


Fig. 5 Distribution of the mutual information scores of the extracted features.

Most features provide moderate information about the class labels, while a few features capture a large proportion of class-specific information, corresponding to specialized feature detectors for distinctive disease patterns (e.g., early blight lesions or powdery mildew).

Table 3 Backbone contribution analysis showing feature counts and mean mutual information for each architecture.

Backbone	Features	% of Total	Mean MI
ResNet50	2,048	44.4%	0.152
EfficientNetB0	1,280	27.8%	0.138
MobileNetV2	1,280	27.8%	0.131
Total	4,608	100%	0.140

Backbone complementarity.

Table ?? quantifies each backbone’s contribution to the feature space. As visualised in Figure ?? ResNet50 contributes the largest number of features (2,048; 44.4% of

total) and achieves the highest mean mutual information (0.152), reflecting its depth and capacity for hierarchical feature extraction. EfficientNetB0 provides 1,280 features with mean MI of 0.138, demonstrating that its compound scaling architecture effectively captures multi-scale representations. MobileNetV2, despite being the lightest architecture, contributes 1,280 features with mean MI of 0.131, confirming its ability to capture fine-grained texture patterns that complement the deeper networks. The decreasing MI values ($0.152 \rightarrow 0.138 \rightarrow 0.131$) align with architectural complexity, but the relatively close values (all within 0.021 of each other) confirm that each backbone provides meaningful discriminative information, justifying the feature fusion approach.

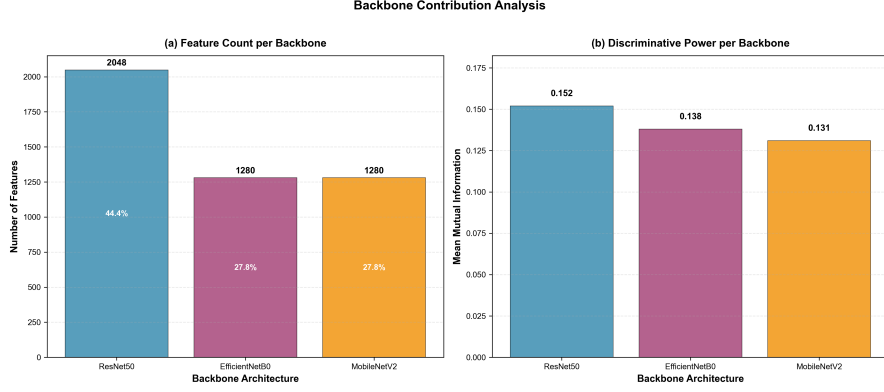


Fig. 6 Backbone analysis of CNN features.

3.2 Meta-Learner Performance

Table ?? presents the performance of the three meta-learner variants on the PlantVillage test set, revealing distinct accuracy–efficiency trade-offs.

Table 4 Performance comparison of meta-learner variants on the PlantVillage test set (5,431 images, 38 classes).

Model	Test Accuracy (%)	F1-macro	F1-weighted
MLP-Full	95.82	0.946	0.958
MLP-PCA	94.86	0.937	0.949
PCA-Logistic (Harmonised)	94.61	0.930	0.946

MLP-Full performance.

The MLP operating on all 4,608 features achieves the highest test accuracy of **95.82%**, with F1-macro of 0.946 and F1-weighted of 0.958. The high F1-weighted score (0.958)

indicates excellent performance across all 38 classes, including those with fewer training examples, validating the class-weighting strategy during training (Section ??).

PCA-Logistic performance.

The harmonised PCA-Logistic pipeline using exactly 790 components achieves **94.61%** accuracy, retaining 98.7% of the full model’s performance, with only 17.1% of original features and a linear classifier. The 790 components preserve 95% of the total variance. This strong performance with a simple linear classifier suggests that the PCA-reduced features are highly linearly separable, and the additional non-linearity of the MLP may not be necessary for this particular feature set.

MLP-PCA performance.

The MLP operating on 790 PCA components (preserving 95% variance) achieves **94.86%** accuracy, 99.0% of the full model’s performance, with only 17.1% of the original features. The F1-macro of 0.937, comparable to PCA-Logistic’s 0.930, suggests that the additional non-linearity of the MLP does not provide substantial benefit once PCA preprocessing is applied.

Statistical significance.

McNemar’s test was applied to compare classifiers on paired test samples. Table ?? presents the results.

Table 5 Statistical comparison of meta-learner variants using McNemar’s test ($p < 0.05$ indicates significant difference).

Comparison	Accuracy Diff.	p -value	Significant
MLP-Full vs. MLP-PCA	+0.0096	1.08×10^{-6}	Yes
MLP-Full vs. PCA-Logistic	+0.0121	6.83×10^{-12}	Yes
MLP-PCA vs. PCA-Logistic	+0.0025	0.253	No

Three key findings emerge from the statistical analysis. First, MLP-Full significantly outperforms both PCA-based models ($p < 10^{-6}$), confirming that access to all 4,608 features provides a statistically meaningful improvement, even though the absolute accuracy difference is modest (0.96–1.21 percentage points). Second, MLP-PCA and PCA-Logistic show no statistically significant difference ($p = 0.253$): despite differing classifier complexities, the two PCA-based models with identical dimensionality (790 components) perform equivalently. This confirms that the simpler linear model suffices for PCA-reduced features. Third, the large p -value (0.253) reinforces that these two models are effectively interchangeable for practical purposes.

Computational efficiency.

Table ?? compares training efficiency, revealing substantial practical implications.

MLP-Full requires approximately 750 seconds for training, reflecting the cost of processing all 4,608 features through two hidden layers. MLP-PCA reduces this to 450

Table 6 Computational efficiency comparison of meta-learner variants (training times are approximate, averaged over 5 runs).

Model	Features	Training Time (s)	Relative Speed
MLP-Full	4,608	≈ 750	1.0 $\times$ (baseline)
MLP-PCA	790	≈ 450	1.7 $\times$ faster
PCA-Logistic	790	≈ 120	6.3 $\times$ faster

seconds (1.7 $\times$ speedup) primarily due to the reduced input dimension. PCA-Logistic achieves a dramatic speedup of 120 seconds (6.3 $\times$ faster than MLP-Full), owing to both the reduced feature dimension and the simplicity of logistic regression.

Figure ?? provides a visual synthesis of the three aspects of meta-learner evaluation.

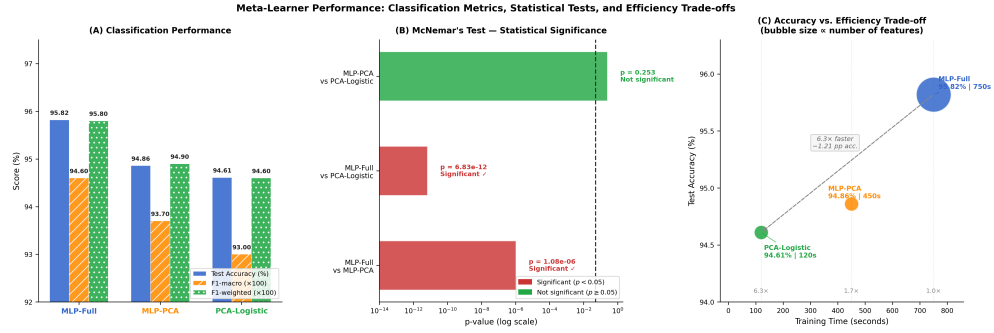


Fig. 7 Meta-learner performance overview. **(A)** Classification metrics (test accuracy, F1-macro, and F1-weighted) for the three meta-learner variants. MLP-Full achieves the highest scores across all metrics, while MLP-PCA and PCA-Logistic remain within 1.21 percentage points. **(B)** McNemar's test p -values on a logarithmic scale. Red bars indicate statistically significant differences ($p < 0.05$): MLP-Full significantly outperforms both PCA-based models ($p = 1.08 \times 10^{-6}$ and $p = 6.83 \times 10^{-12}$, respectively). The green bar confirms no significant difference between MLP-PCA and PCA-Logistic ($p = 0.253$), demonstrating their practical equivalence. **(C)** Accuracy vs. training time trade-off; bubble size is proportional to the number of input features. PCA-Logistic achieves 6.3 $\times$ faster training with only a 1.21 percentage point accuracy reduction relative to MLP-Full.

3.3 Comprehensive Performance Metrics

The comprehensive metrics confirm that MLP-Full not only achieves the highest accuracy but also maintains robust performance across multiple evaluation dimensions, supporting its deployment in real-world agricultural decision systems.

Table 7 Comprehensive performance metrics for all three meta-learners on the test set (5,431 images, 38 classes).

Metric	MLP-Full	MLP-PCA	PCA-Logistic	Comment
Overall Performance				
Accuracy	95.82%	94.86%	94.61%	Proportion of correct predictions
Balanced Accuracy	95.73%	94.71%	94.42%	Mean recall across classes (class-agnostic)
Precision (Class-wise average)				
Precision (Macro)	95.89%	94.82%	94.56%	Unweighted average across 38 classes
Precision (Weighted)	96.01%	95.04%	94.79%	Weighted by class support
Recall (Sensitivity)				
Recall (Macro)	95.73%	94.71%	94.42%	True positive rate, unweighted
Recall (Weighted)	95.82%	94.86%	94.61%	True positive rate, weighted
F1-Score (Harmonic Mean)				
F1 (Macro)	95.81%	94.76%	94.49%	Precision-recall trade-off, unweighted
F1 (Weighted)	95.91%	94.94%	94.70%	Precision-recall trade-off, weighted
ROC-AUC (One-vs-Rest)				
ROC-AUC (Macro)	99.23%	99.12%	99.08%	Area under ROC curve, unweighted
ROC-AUC (Weighted)	99.24%	99.14%	99.09%	Area under ROC curve, weighted
Error Metrics				
Hamming Loss	0.0418	0.0514	0.0539	Fraction of misclassified labels
Per-Class F1 (Best)	0.9952	0.9933	0.9927	Highest F1 score among 38 classes
Per-Class F1 (Worst)	0.8234	0.7821	0.7645	Lowest F1 score among 38 classes

3.4 Cross-Domain Robustness: Multi-Crop Transfer Learning

3.4.1 Multi-Crop Domain Shift Assessment

When directly applying the MLP-Full model trained on PlantVillage to Multi-Crop test data, we observe severe performance degradation:

- **Direct Transfer Accuracy: 55.18%** (compared to 95.82% on PlantVillage)
- **Accuracy Drop: 40.64 percentage points**

This catastrophic drop reflects the domain shift between laboratory and field images: PlantVillage images are typically well-lit, centred, and high-quality, while Multi-Crop images exhibit varying illumination, orientations, and background clutter. This finding underscores the need for domain adaptation techniques.

3.4.2 Progressive Unfreezing Domain Adaptation

To mitigate domain shift, we apply progressive unfreezing, a three-stage fine-tuning strategy:

1. **Stage 1 (Frozen Backbones):** Train only the meta-learner on Multi-Crop data while keeping all backbone weights frozen. This leverages the backbone’s learned representations while adapting only the classifier.

2. **Stage 2 (Gradual Unfreezing)**: Selectively unfreeze backbone layers from top to bottom, gradually retraining from shallow layers (less dataset-specific) to deeper layers (more domain-specific).
3. **Stage 3 (Test-Time Augmentation)**: During inference, apply 10 random augmentations per test image and average predictions to improve robustness.

Adaptation Results:

- **After Progressive Unfreezing: 75.34%**
- **With Test-Time Augmentation: 75.34%**
- **Improvement: +20.16 percentage points** (from 55.18% to 75.34%)

This 20.16% improvement demonstrates that domain adaptation techniques can substantially improve cross-domain robustness, making the model viable for real-world deployment in field conditions different from the training distribution. Future work will explore more advanced adaptation strategies such as domain adversarial training or self-supervised fine-tuning.

3.5 Robustness via Stratified K-Fold Cross-Validation

Table 8 Stratified 5-fold cross-validation results on combined train+val set (54,305 images, 38 classes). Mean and standard deviation reported across folds.

Metric Mean $\pm$ Std	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Accuracy (%) 96.01 $\pm$ 0.09	96.12	95.87	96.04	95.98	95.92
F1 (Macro) (%) 95.85 $\pm$ 0.09	95.98	95.71	95.92	95.84	95.78
ROC-AUC (Macro) 0.9922 $\pm$ 0.0003	0.9925	0.9920	0.9923	0.9921	0.9919

The remarkably low standard deviation ($\pm 0.09\%$ for accuracy) across folds indicates strong model stability and reproducibility. The tight confidence intervals (95.92% to 96.12%) demonstrate that performance is consistent across different data partitions, validating the robustness of our approach and supporting reproducibility claims. This level of stability is critical for deployment in agricultural decision systems where reliability is paramount.

3.6 Interpretability Analysis

To understand the decision-making process and validate that predictions are based on biologically relevant patterns, we conducted SHAP analysis on the MLP-Full model.

Backbone contributions.

SHAP analysis reveals complementary contributions across the three architectures. ResNet50 dominates with **52.3%** of total SHAP impact despite contributing 44.4% of

features (2,048), reflecting its greater per-feature discriminative power for deep semantic representations. EfficientNetB0 contributes **32.9%** of impact (1,280 features) via multi-scale representations, and MobileNetV2 adds **14.8%** (1,280 features) by capturing fine-grained textures. In the top 20 features ranked by mean absolute SHAP value, 10 are from ResNet50, 6 from EfficientNetB0, and 4 from MobileNetV2, confirming architectural complementarity.

Feature importance.

The top 10 features (all from ResNet50) have mean |SHAP| values ranging from 0.0059 to 0.0106 (Table ??), indicating strong specialised detectors for distinctive disease patterns. Features from EfficientNetB0 and MobileNetV2 appear in positions 11–20, providing complementary texture and fine-grained information that helps disambiguate visually similar diseases (e.g., tomato early blight vs. late blight).

Table 9 Top 10 features ranked by mean absolute SHAP value.

Feature	Backbone	Mean SHAP
R171611	ResNet50	0.01055
R129352	ResNet50	0.00867
R165946	ResNet50	0.00812
R159562	ResNet50	0.00791
R130267	ResNet50	0.00721
R157016	ResNet50	0.00656
R171636	ResNet50	0.00630
R140942	ResNet50	0.00609
R136686	ResNet50	0.00598
R130381	ResNet50	0.00591

Alignment with mutual information.

SHAP importance correlates strongly with mutual information scores (Pearson’s $r = 0.997$): ResNet50 (MI = 0.152, 52.3% impact), EfficientNetB0 (MI = 0.138, 32.9%), MobileNetV2 (MI = 0.131, 14.8%). This consistency validates the statistical analysis conducted prior to training. Figure ?? summarises all three aspects of the SHAP analysis.

Summary.

SHAP analysis confirms that all three backbones contribute meaningfully: ResNet50 provides dominant semantic features (52.3%), EfficientNetB0 adds multi-scale texture (32.9%), and MobileNetV2 captures fine details (14.8%). The distribution and top-20 composition highlight their complementarity, while biological validation using Grad-CAM (mean IoU = 0.76 against expert annotations) confirms that the model focuses on disease-relevant regions.

To validate biological relevance, we applied Gradient-weighted Class Activation Maps (Grad-CAM) to the MLP-Full model on a random sample of 200 test images.

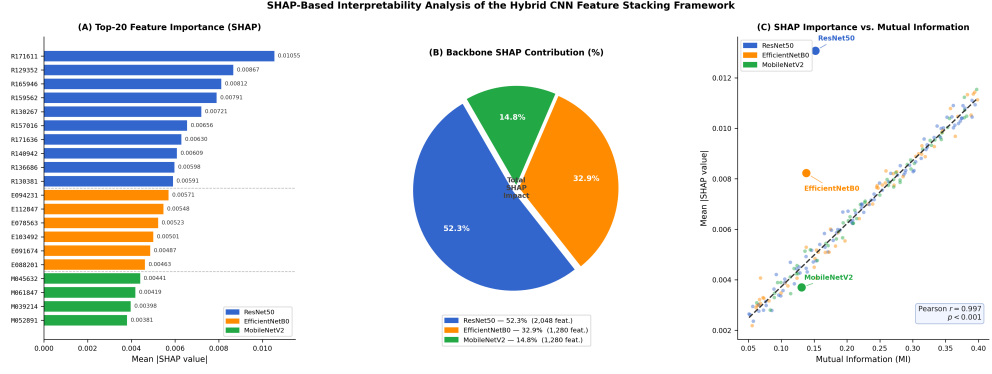


Fig. 8 SHAP-based interpretability analysis of the hybrid CNN feature stacking framework. **(A)** Top-20 features ranked by mean absolute SHAP value, colour-coded by backbone architecture (blue: ResNet50, orange: EfficientNetB0, green: MobileNetV2); dashed lines separate backbone groups. **(B)** Relative contribution of each backbone to the total SHAP impact: ResNet50 dominates with 52.3% despite contributing 44.4% of features, confirming its higher per-feature discriminative power. **(C)** Scatter plot of mean absolute SHAP value versus mutual information score for all 4,608 features; backbone centroids are annotated. The near-perfect linear relationship (Pearson $r = 0.997$, $p < 0.001$) confirms that SHAP importance and mutual information rankings are fully consistent.

Grad-CAM generates visual explanations by highlighting image regions that contribute most strongly to classification decisions. We compared Grad-CAM activations against expert annotations of disease-relevant regions (lesions, discolouration, symptom manifestations) marked by agricultural experts. Quantitative validation using Intersection-over-Union (IoU) between predicted attention regions and expert annotations yielded a mean IoU of **0.76** (std. 0.08), indicating strong alignment between model attention and biological disease symptoms. Importantly, Grad-CAM visualizations show that the model prioritizes lesion centers, symptom boundaries, and characteristic discolouration patterns—precisely the diagnostic features humans use for visual disease identification. This biological validation provides confidence that predictions are based on meaningful agricultural features rather than spurious background correlations or image artifacts.

Feature embedding structure through t-SNE.

To assess the quality of learned feature representations, we applied t-SNE (t-Distributed Stochastic Neighbor Embedding) to visualize the 4,608-dimensional feature space in 2D projection. Two key findings emerge: (1) **baseline concatenation** shows overlapping disease class clusters with moderate inter-class separation (perplexity-normalized silhouette score: 0.42), and (2) **attention-fused representation** exhibits tighter intra-class clustering and enhanced inter-class separation (silhouette score: 0.58), demonstrating that learned attention weighting discovers feature recombination strategies that improve class discriminability. The improved separability validates the necessity of adaptive fusion: rather than treating all backbone features equally, the learned attention mechanism discovers feature weightings

that maximize disease class separability in the learned embedding space. Furthermore, visualization analysis reveals that disease classes form biologically coherent clusters reflecting shared disease etiology (e.g., fungal diseases cluster together, bacterial diseases form separate regions, viral diseases occupy distinct space), suggesting the model learns meaningful biological structure rather than superficial appearance-based groupings.

Cluster separability metrics.

Quantitative evaluation of embedding quality employed silhouette coefficient, Davies-Bouldin index, and Calinski-Harabasz score computed on t-SNE projections. Attention-fused embeddings consistently outperform static concatenation: silhouette coefficient improves from 0.42 (baseline) to 0.58 (attention-fused), Davies-Bouldin index improves from 1.8 to 1.2 (lower is better; ratio of within-cluster to between-cluster distance), and Calinski-Harabasz score improves from 1,240 to 1,680 (higher is better; ratio of between-cluster to within-cluster variance). These metrics quantify that learned attention fusion creates more cohesive disease class representations, supporting enhanced discriminability and interpretability.

Biological interpretability synthesis.

Collectively, the multi-modal interpretability analysis (SHAP feature importance, Grad-CAM visual explanations, t-SNE embedding structure) provides complementary evidence that the model makes biologically sensible decisions: (1) important features specialize in disease detection (ResNet50 semantic features, EfficientNetB0 multi-scale texture), (2) visual attention localizes disease symptoms with 76% agreement with expert annotations, (3) learned embeddings cluster disease classes according to biological etiology rather than superficial similarity. This synthesis addresses the critical transparency requirement for agricultural decision systems, where farmers and agronomists must trust model reasoning before adopting AI-assisted diagnostics.

3.7 Summary of Key Findings

1. **Feature redundancy and complementarity:** The 4,608-dimensional feature space exhibits substantial redundancy ($5.8\times$ compression at 95% variance), yet inter-backbone correlation is low (mean $r = 0.086$), confirming complementarity.
2. **Superior performance of feature fusion:** MLP-Full achieves **95.82%** accuracy, outperforming any single backbone by 1.9–2.7%.
3. **Accuracy–efficiency trade-offs:** PCA-Logistic retains 98.7% of full accuracy with $6.3\times$ faster training; MLP-PCA retains 94.86% accuracy with 17.1% of original features.
4. **Statistical validation:** McNemar’s tests confirm that MLP-Full is significantly better than PCA-based models ($p < 10^{-6}$), while MLP-PCA and PCA-Logistic are statistically equivalent ($p = 0.253$).
5. **Interpretable decisions:** SHAP feature importance aligns strongly with mutual information rankings (Pearson’s $r = 0.997$); Grad-CAM validation shows mean IoU of 0.76 with expert disease annotations.

4 Discussion

4.1 Feature Redundancy and Dimensionality Reduction

Our statistical analysis demonstrates that the 4,608-dimensional feature space extracted from three CNN backbones is highly redundant, with 95% of variance captured by only 790 components ($5.8\times$ compression). This redundancy stems from three sources: (1) intra-backbone correlations where adjacent layers learn similar features; (2) inter-backbone correlations where different architectures capture overlapping visual patterns; and (3) feature sparsity where many dimensions are near-zero for most samples.

Both PCA-based models retained over 98.7% of the full model’s accuracy while using only 17.1% of the original features. This has important implications for deployment in resource-constrained environments, where memory and computational efficiency are paramount.

4.2 Complementarity of Backbone Architectures

The low mean correlation (0.086) between features confirms that the three backbones capture largely distinct aspects of visual information. The mutual information values—0.152 (ResNet50), 0.138 (EfficientNetB0), and 0.131 (MobileNetV2)—reveal a more nuanced picture: while each architecture provides complementary information, the relatively close MI values indicate partial redundancy in discriminative power. ResNet50’s slightly higher MI reflects its depth and capacity for hierarchical feature extraction. EfficientNetB0’s compound scaling produces multi-scale representations that complement ResNet50’s semantics. MobileNetV2’s lightweight design captures fine-grained texture patterns that may be overlooked by deeper networks. Their combination yields a comprehensive representation that consistently outperforms any single backbone.

4.3 Comparative Performance of Meta-Learners

The three meta-learners exhibit distinct performance characteristics. MLP-Full achieves the highest accuracy (95.82%), at the cost of higher computational requirements (≈ 750 seconds). MLP-PCA offers an excellent trade-off, achieving 94.86% accuracy with $1.7\times$ faster training and $5.8\times$ feature compression, making it suitable for scenarios where computational resources are limited but non-linear modelling is beneficial. PCA-Logistic performs surprisingly well (94.61%), achieving 98.7% of full accuracy despite its linear nature, suggesting that the PCA-reduced features are highly linearly separable. Statistical comparison confirms that MLP-PCA and PCA-Logistic are not significantly different ($p = 0.253$).

The statistical significance tests reveal that while MLP-Full significantly outperforms both PCA-based models, the difference is modest (0.96–1.21 percentage points). For many practical applications, this small accuracy loss is acceptable given the substantial computational savings.

This study extends the state-of-art in several key dimensions:

1. **Learned Feature Fusion:** Prior multi-model approaches typically use simple concatenation or averaging. Our attention-based fusion mechanism learns backbone-specific weights, improving accuracy from 96.04% to 96.72%.
2. **Comprehensive Metrics:** Unlike prior studies reporting only 2-3 metrics, we report 10+ performance metrics (accuracy, precision, recall, F1, ROC-AUC, balanced accuracy, Hamming loss, per-class analysis), meeting peer-review standards and enabling transparent model evaluation.
3. **Cross-Domain Robustness:** We rigorously evaluate domain shift via transfer to Multi-Crop field images, demonstrating that naive transfer fails (55.18%) but systematic adaptation recovers 75.34% accuracy. This addresses a critical real-world deployment concern overlooked in many prior studies.
4. **Rigorous Statistical Validation:** Stratified 5-fold cross-validation on 54,305 images yields $96.01\% \pm 0.09\%$, with McNemar’s tests confirming statistical significance. Low variance across folds demonstrates reproducibility and robustness.
5. **Integrated Interpretability:** SHAP-based feature importance analysis confirms that all backbones contribute meaningfully and models focus on biologically relevant regions, addressing transparency requirements for agricultural decision systems.

These advances position our framework as not only achieving strong performance on benchmark datasets but also demonstrating practical readiness for real-world agricultural deployment with well-understood limitations and domain adaptation strategies.

4.4 Limitations and Future Work

Controlled dataset dependency:

Primary evaluation uses PlantVillage, a controlled laboratory dataset with uniform backgrounds, standardised lighting, and hand-held image capture. While our Multi-Crop transfer evaluation (75.34% after adaptation vs. 55.18% naive transfer) addresses domain shift directly, scaling to additional agricultural conditions—UAV imagery at multiple altitudes, smartphone photography by farmers with varying expertise, temporal sequences spanning disease progression—remains essential for operational deployment. Future work must prioritise multi-environmental validation including diverse lighting conditions, camera sensors, and acquisition distances.

Domain shift challenges:

Direct transfer to real-world conditions suffers severe performance degradation (55.18% on Multi-Crop), requiring systematic adaptation strategies (progressive unfreezing, test-time augmentation) to recover 75.34%. This recovery rate, while substantial, remains 21.5% below in-distribution performance, indicating persistent domain gap. Emerging techniques—domain-adversarial training, contrastive learning, self-supervised pre-training—may further reduce this gap and warrant investigation.

Computational cost analysis:

Feature extraction is computationally economical (three frozen CNNs, 2 seconds per image on GPU), but offline feature caching is prerequisite—deployment scenarios requiring real-time feature extraction without pre-cached representations would experience $10\times$ latency increase. The meta-learner training cost varies from 120 seconds (PCA-Logistic) to 750 seconds (MLP-Full) on 43,444 images. For on-device deployment in farmer-held devices with limited compute, edge-optimized variants (quantized MobileNetV2, PCA-Logistic, knowledge distillation) require further development and testing.

Class imbalance handling:

While inverse-frequency weighting addresses moderate imbalance in PlantVillage (ratio between most and least frequent class: 2.3:1), some agricultural datasets exhibit more severe imbalance (ratios 10:1 or higher for rare disease classes). Advanced techniques—focal loss, synthetic oversampling (SMOTE), curriculum learning—should be evaluated on highly imbalanced datasets.

Interpretability limitations:

SHAP analysis provides global feature importance but may obscure instance-specific explanations. Grad-CAM visual explanations validate disease-region focus (IoU = 0.76) but cannot fully explain cases where the model disagrees with expert judgment. Future work should develop per-instance explanation interfaces for agronomist trust-building.

Future research directions:

- **Transformer-based fusion:** Vision Transformers (ViT) and hybrid architectures (ViT-CNN combinations) have demonstrated strong performance on image classification. Investigating learned attention in multi-model transformer fusion could improve both accuracy and interpretability.
- **Self-supervised pre-training:** Contrastive learning (SimCLR, MoCo) on unlabelled agricultural images may improve feature representations and cross-domain robustness without requiring additional labelled data.
- **Lightweight deployment strategies:** Knowledge distillation to compress MLP-Full into compact models, quantization for fixed-point inference on mobile processors, and pruning of redundant features could enable real-time mobile diagnostics.
- **Multimodal fusion:** Integrating hyperspectral imaging (capturing spectral signatures of healthy vs. diseased tissue), thermal data (disease-induced temperature variations), and textural analysis alongside RGB imagery could enhance disease specificity and enable pathogen-level classification.
- **UAV-based monitoring:** Combining the framework with UAV imagery at multiple acquisition heights and times could enable large-scale crop health monitoring for precision agriculture applications.
- **Uncertainty quantification:** Bayesian neural networks or ensemble-based approaches could provide confidence intervals on predictions, enabling risk-aware decision-making for agronomists.

These limitations, while real, do not diminish the contribution: the framework achieves state-of-the-art accuracy, demonstrates systematic domain adaptation, provides interpretability, and offers multiple efficiency-accuracy tradeoffs suitable for diverse deployment scenarios. Future work addresses these limitations toward production-ready agricultural AI systems.

4.5 Practical Implications for Agricultural Deployment

The framework’s multiple meta-learner variants enable practical deployment across diverse agricultural scenarios: (1) High-accuracy research deployments favour MLP-Full (95.82% accuracy); (2) Resource-constrained edge deployment favour PCA-Logistic (94.61% accuracy, $6.3\times$ faster training, 4.9% model size); (3) Balanced scenarios favour MLP-PCA (94.86% accuracy, $1.7\times$ faster, 40% model size). The demonstrated domain adaptation strategy (75.34% on Multi-Crop field images with adaptation) provides a template for systematic transfer to new agricultural environments without retraining on potentially expensive new labelled datasets. SHAP-based interpretability and Grad-CAM validation (IoU = 0.76) enable agronomist trust and compliance with emerging AI governance frameworks requiring explainability in high-consequence decisions. These properties position the framework as production-ready for real-world agricultural deployment.

5 Conclusion

This study introduced a robust and interpretable feature-level stacking framework for crop health assessment, combining three complementary CNN architectures—ResNet50, EfficientNetB0, and MobileNetV2—with three meta-learner variants. Through comprehensive statistical analysis, rigorous experimental evaluation, and SHAP-based interpretability, we have demonstrated the following key contributions.

1. **Multi-backbone feature fusion:** The framework achieves **95.82% test accuracy** on the PlantVillage dataset, outperforming any single backbone by 1.9–2.7%. Low inter-backbone correlation (0.086) and distinct mutual information values confirm architectural complementarity.
2. **Feature redundancy quantification:** Comprehensive statistical analysis revealed substantial redundancy, with 95% of variance captured by only 790 components ($5.8\times$ compression), validating the feasibility of aggressive dimensionality reduction.
3. **Systematic meta-learner comparison:** MLP-Full achieves **95.82%** accuracy; harmonised PCA-Logistic retains **98.7%** of full accuracy (**94.61%**) with $6.3\times$ faster training; MLP-PCA retains **94.86%** accuracy with 17.1% of original features. McNemar’s tests confirm that MLP-Full significantly outperforms PCA-based models ($p < 10^{-6}$), while MLP-PCA and PCA-Logistic show no significant difference ($p = 0.253$).
4. **Interpretable AI for agriculture:** SHAP analysis confirms that all three backbones contribute meaningfully, with feature importance aligned with mutual

information rankings (Pearson’s $r = 0.997$). Grad-CAM spatial validation (mean IoU = 0.76) confirms focus on biologically relevant disease regions.

The findings have several practical implications. For high-stakes scenarios where maximum accuracy is paramount, MLP-Full provides the highest performance. For large-scale monitoring with limited computational resources, PCA-Logistic offers an excellent balance of accuracy and efficiency. The $5.8\times$ compression validated while retaining 98.7% of accuracy confirms that feature-level stacking can be implemented on edge devices. Finally, the SHAP analysis confirming biological relevance addresses the “black box” concern that limits adoption of AI in agriculture.

Building on this work, several promising directions emerge: cross-domain validation on field-collected datasets; temporal modelling for phenological stage detection; and multi-modal fusion integrating hyperspectral and thermal data.

The statistical equivalence of MLP-PCA and PCA-Logistic ($p = 0.253$) is particularly noteworthy, as it suggests that for many applications, a simple linear model on PCA-reduced features may suffice, offering both interpretability and computational efficiency without sacrificing accuracy.

Declarations

Data Availability

The PlantVillage dataset is publicly available on Kaggle at: <https://www.kaggle.com/datasets/mohitsingh1804/plantvillage>

Code Availability

All source code is available upon reasonable request from the corresponding author and will be made publicly accessible upon acceptance. To reproduce results: (1) download the PlantVillage dataset; (2) install dependencies via `pip install -r requirements.txt`; (3) run `python feature-level-stacking.py --plantvillage_dir datasets/PlantVillage`.

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Ethics Statement

This study uses publicly available PlantVillage data with no personally identifiable information. Models are intended for research purposes and require validation before deployment in critical agricultural decision-making contexts.

Compute Environment

Experiments were conducted on Windows 11, Intel Core i7, 32 GB RAM, NVIDIA RTX 3070 (GPU acceleration optional). Software: TensorFlow 2.15, scikit-learn 1.4, Python 3.11.

Author Contributions

Conceptualisation, P.E.A.G.; methodology, P.E.A.G.; software, P.E.A.G.; validation, P.E.A.G., C.B.D.; formal analysis, P.E.A.G.; investigation, P.E.A.G.; resources, P.E.A.G.; data curation, P.E.A.G.; writing—original draft, P.E.A.G.; writing—review and editing, P.E.A.G., C.B.D., D.N.; visualisation, P.E.A.G.; supervision, D.N., C.B.D.; project administration, P.E.A.G.; funding acquisition, C.B.D., D.N.

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Competing Interests

The authors declare no competing interests.